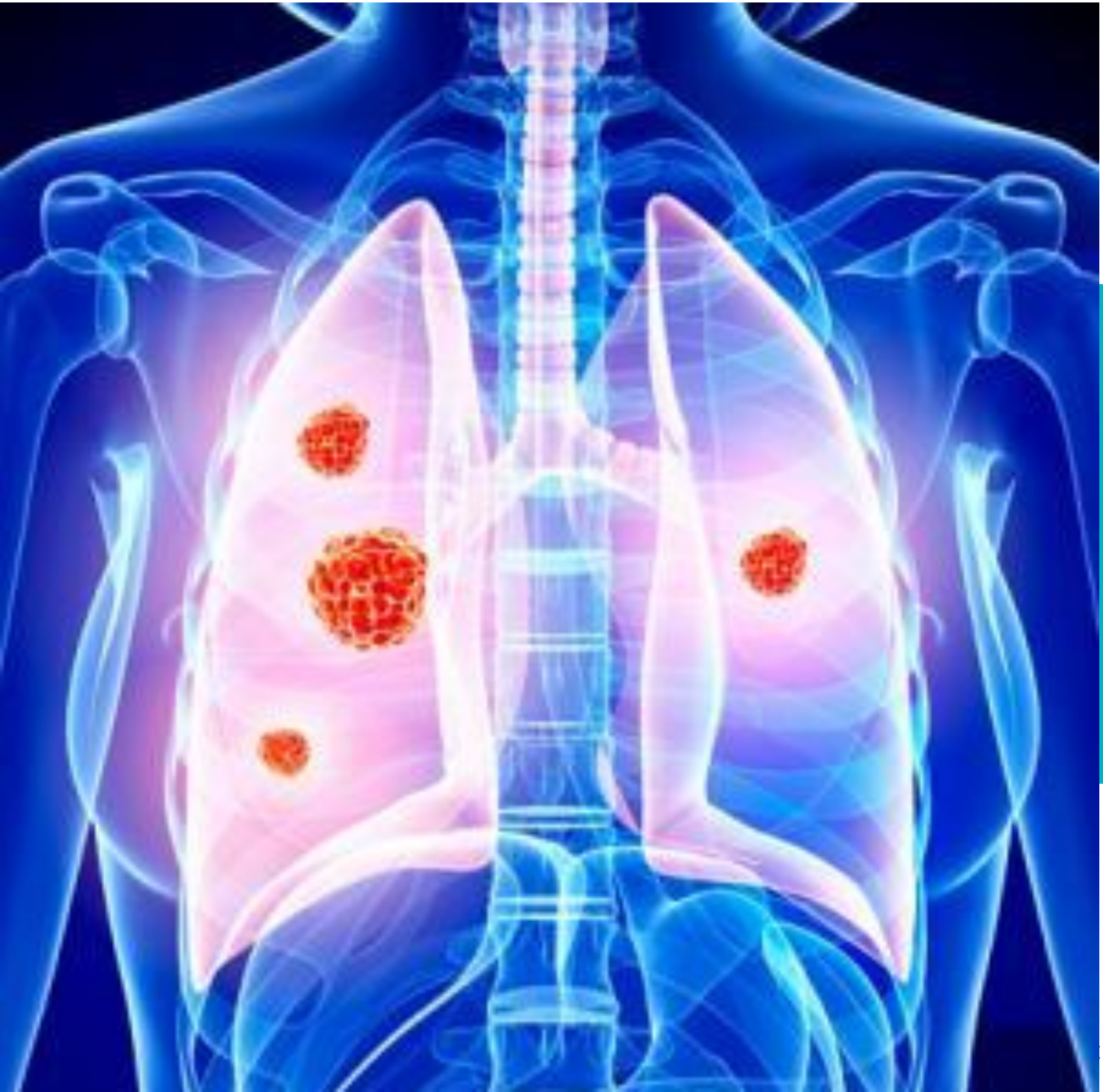
AN INTERPLAY BETWEEN BEHAVIOURAL & PSYCHOSOCIAL FACTORS WITH EARLY CLINICAL MARKERS

LUNG CANCER

Isaac Oselukwue



ABOUT LUNG CANCER

Most frequent cause of cancer death.

Popularity from world war II

Low recovery rates

LITERATURE REVIEW

BEHAVIOURAL RISK FACTORS

Age: 40 to 80 years

Obese: 78% of patients

Smokers: 88% women / 57% men

PSYCHOLOGICAL BURDEN

25% stigma

Post recovery PTSD

Stress from ever smokers

INFLUENCE

Peer pressure

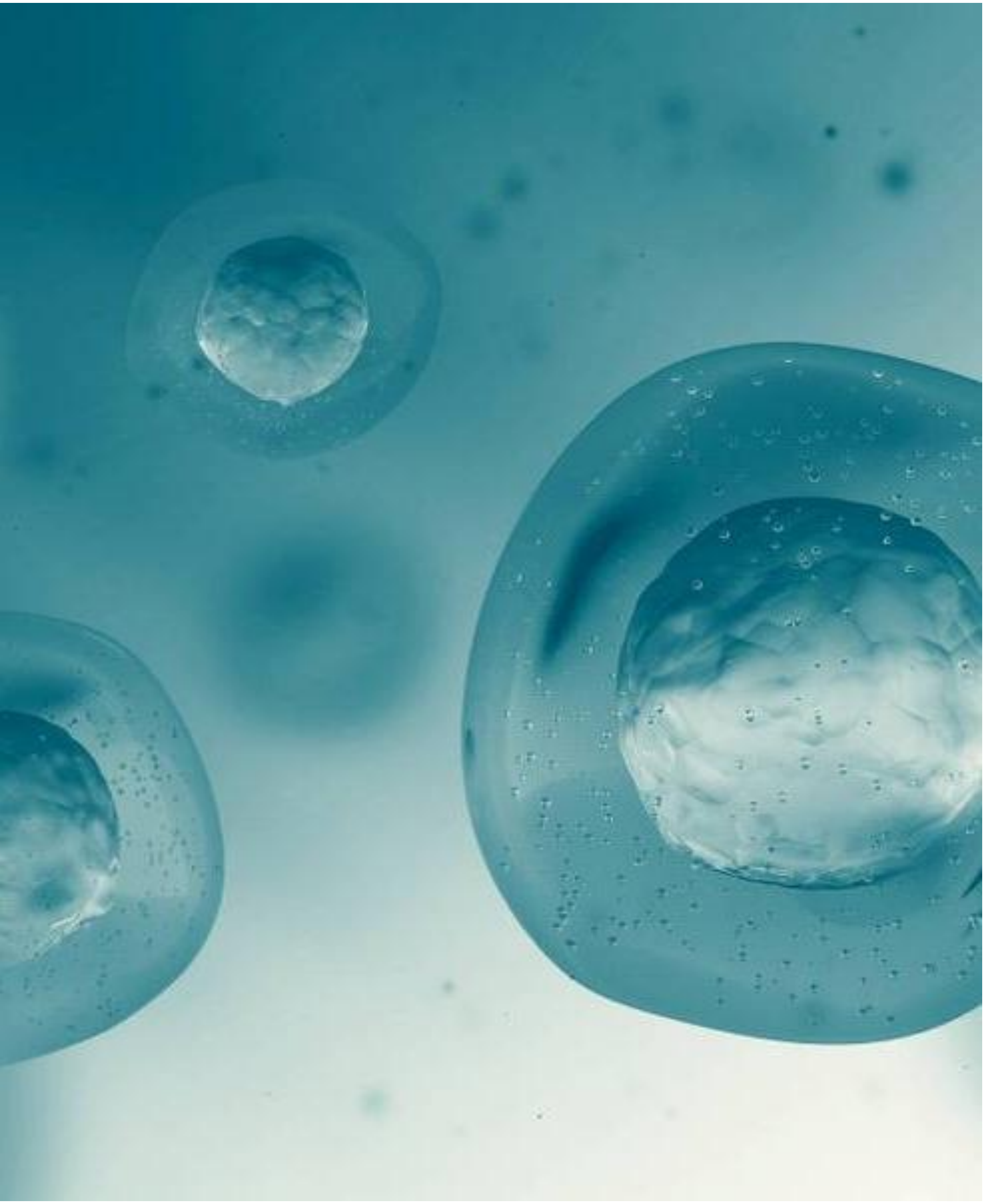
HYPOTHESIS

NULL HYPOTHESIS

There is no significant interaction; the impact of psychosocial and behavioral habits on lung cancer is the sum of their independent events and no integration exists.

ALTERNATIVE HYPOTHESIS

There is a significant interaction; implying that individuals that experience psychosocial stress and behavioral habits have a high risk of lung cancer and this integration is moderated by age and gender



GOAL

QUESTION

Does the interplay between psychosocial factors and behavioral habits increases likelihood of cancer? Is moderated by age and gender?

METHODOLOGY

DATASET DESCRIPTION

Source: Kaggle

License: cco

No PiD (patient identity)

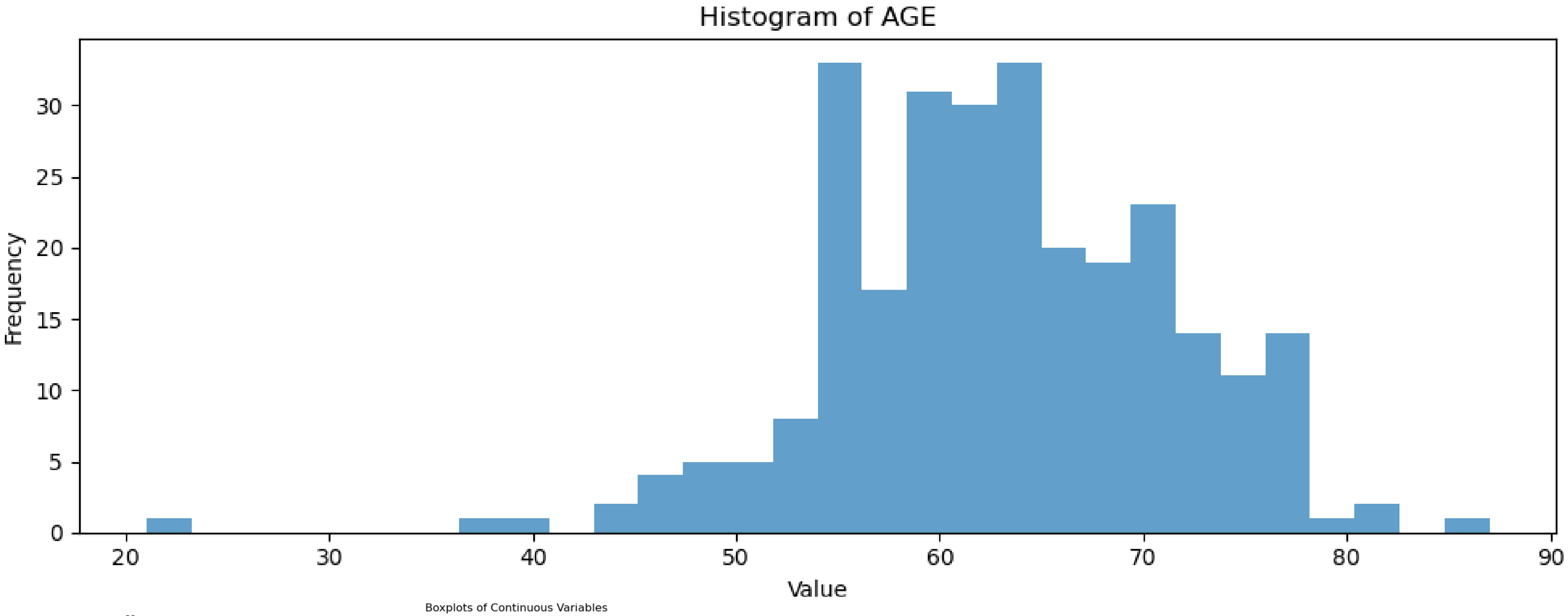
Features: 16

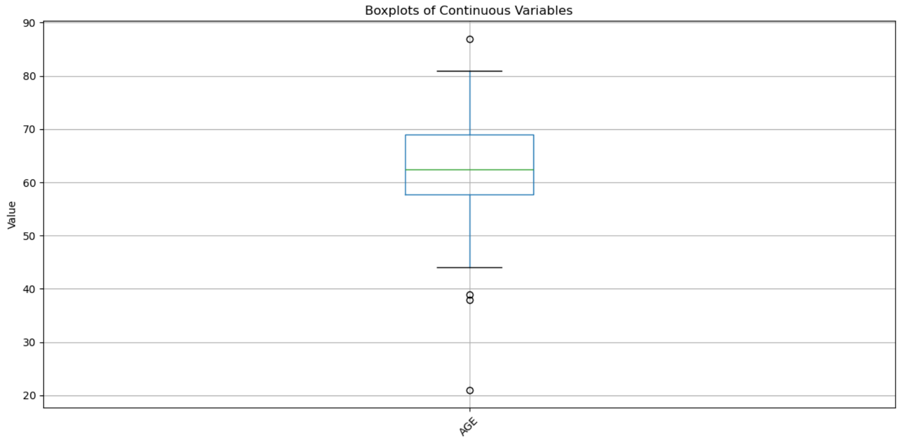
Before: 309 / After: 276

GENDER	AGE	ANXIETY	ALCOHOL CONSUMING	LUNG_CANCER
M	69	2	2	YES

GENDER	AGE	ANXIETY	ALCOHOL CONSUMING	LUNG_CANCER
1	69	1	1	1

STATISTICS





CONTINUOUS VARIABLES

Age

Mean: 62.91

Mode: 64

SD: 8.3

Variance: 70.21

Min Age: 21

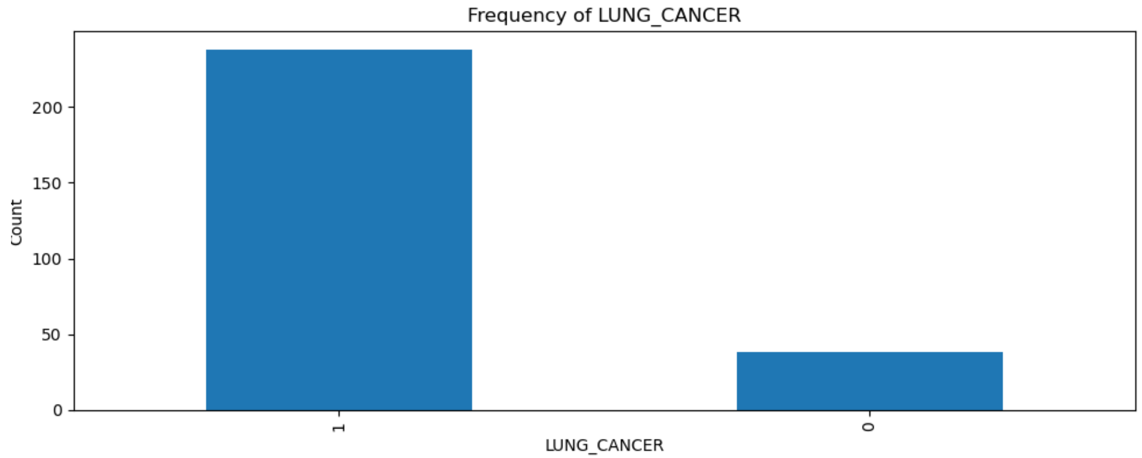
Max Age: 87

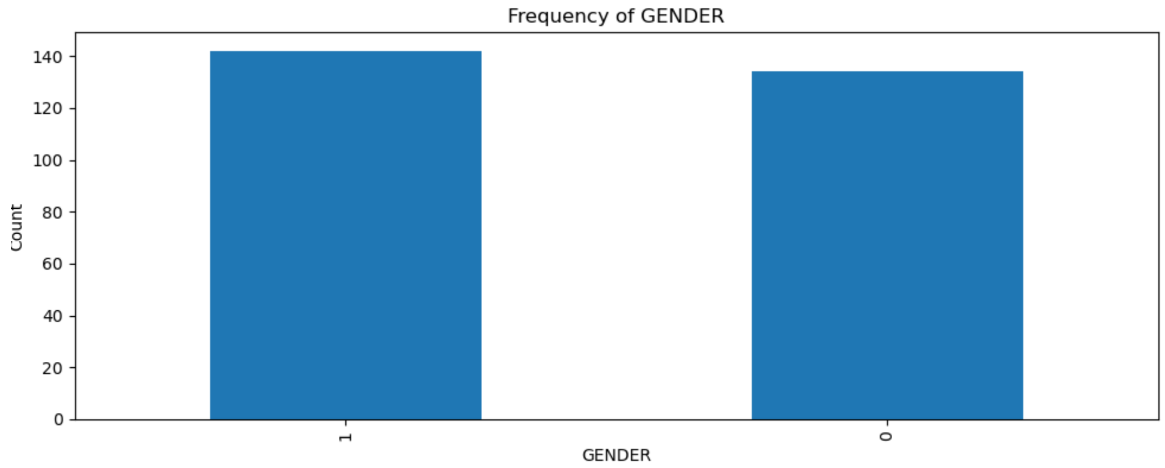
Quartiles: Q1=57.75, Q2=62.50, Q3=69.00

Interquartile Range: 11.25

STATISTICS

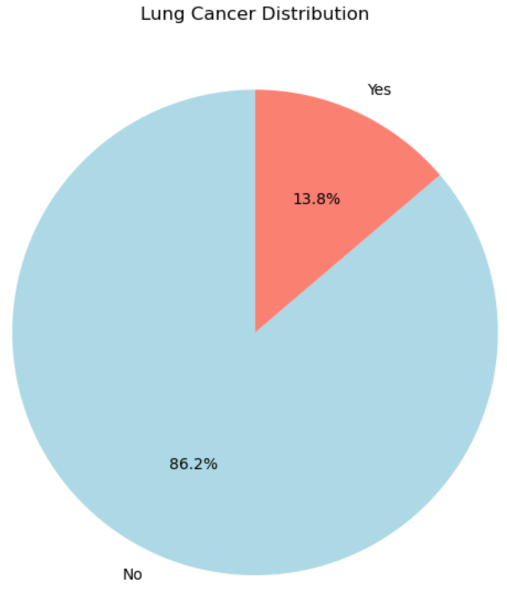
FREQUENCY VARIABLES

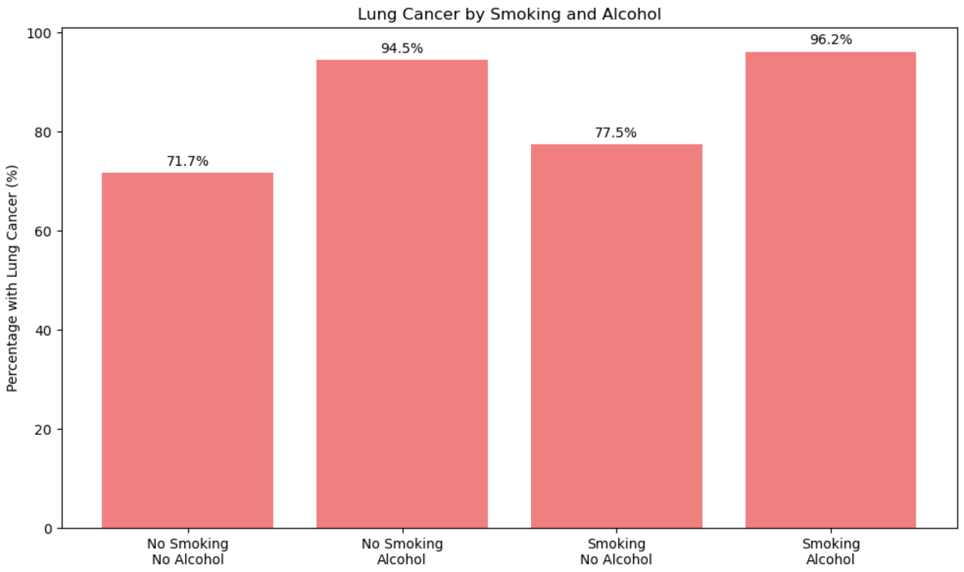




GENDER
| 1 | 142 | 51.45% |
| 0 | 134 | 48.55% |
SMOKING
| 1 | 150 | 54.35% |
| 0 | 126 | 45.65% |
YELLOW_FINGERS
| 1 | 159 | 57.61% |
| 0 | 117 | 42.39% |
ANXIETY
| 0 | 139 | 50.36% |
| 1 | 137 | 49.64% |
PEER_PRESSURE
| 1 | 140 | 50.72% |
| 0 | 136 | 49.28% |
CHRONIC_DISEASE
| 1 | 144 | 52.17% |
| 0 | 132 | 47.83% |
FATIGUE
| 1 | 183 | 66.30% |
| 0 | 93 | 33.70% |
ALLERGY
| 1 | 151 | 54.71% |
| 0 | 125 | 45.29% |

WHEEZING
| 1 | 151 | 54.71% |
| 0 | 125 | 45.29% |
ALCOHOL_CONSUMING
| 1 | 152 | 55.07% |
| 0 | 124 | 44.93% |
COUGHING
| 1 | 159 | 57.61% |
| 0 | 117 | 42.39% |
SHORTNESS_OF_BREATH
| 1 | 174 | 63.04% |
| 0 | 102 | 36.96% |
SWALLOWING_DIFFICULTY
| 0 | 147 | 53.26% |
| 1 | 129 | 46.74% |
CHEST_PAIN
| 1 | 154 | 55.80% |
| 0 | 122 | 44.20% |
LUNG_CANCER
| 1 | 238 | 86.23% |
| 0 | 38 | 13.77% |





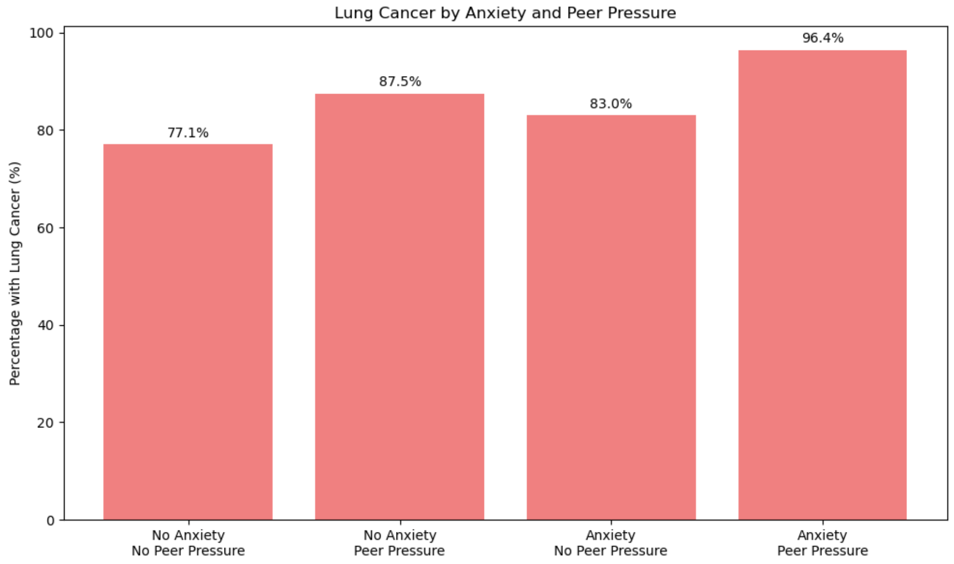
BEHAVIOURAL & LUNG CANCER

96.20% with smoking and alcohol

94.52% with alcohol

77.46% with smoking

71.70% with none



PSYCHOSOCIAL & LUNG CANCER

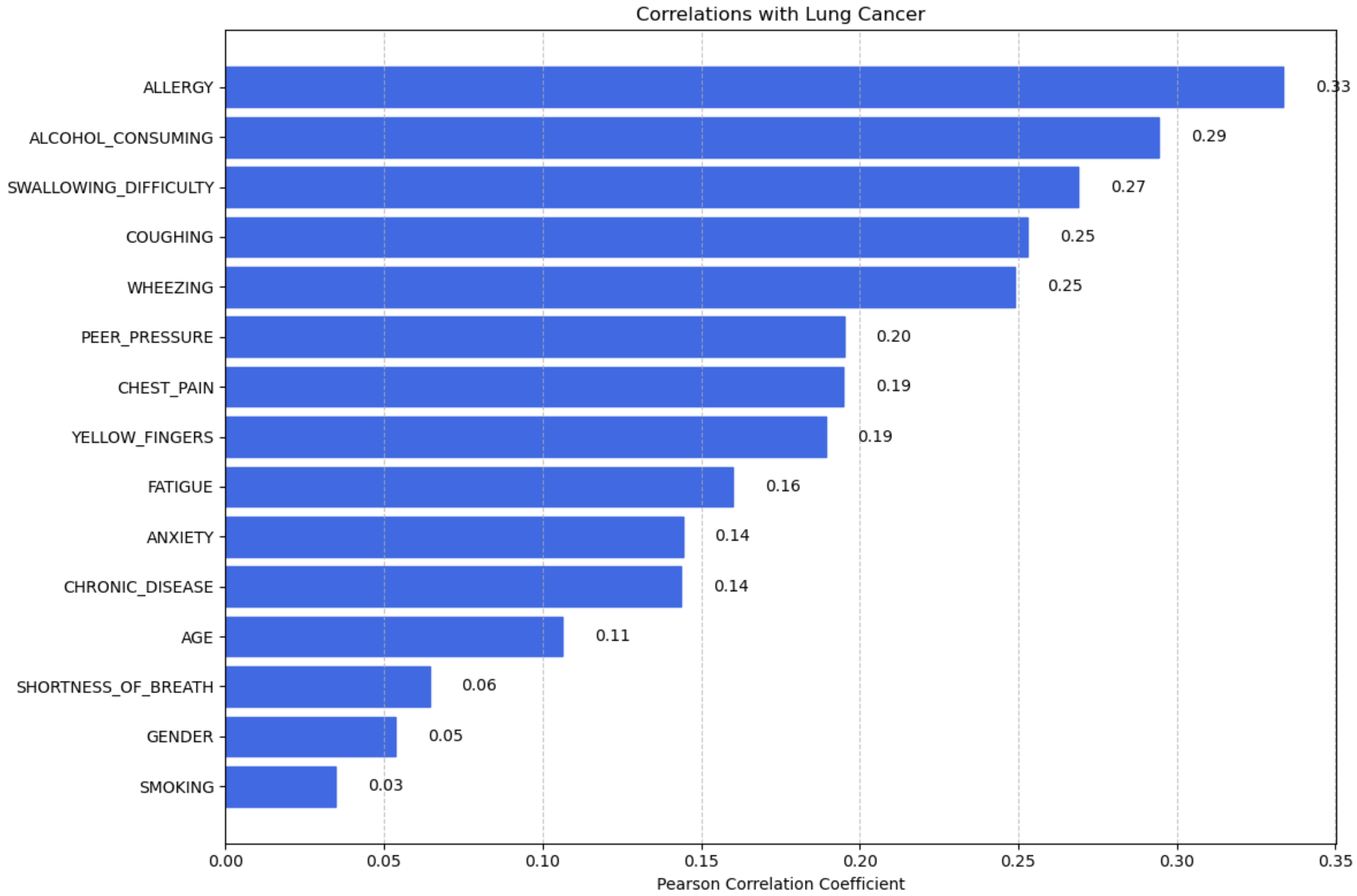
96.7% with Anxiety & Peer pressure

83.02% with Anxiety

87.50% with Peer pressure

77.11% with none

CORRELATION ANALYSIS



TOP POSITIVE

Allergy = 0.33

Alcohol Consumption = 0.2944

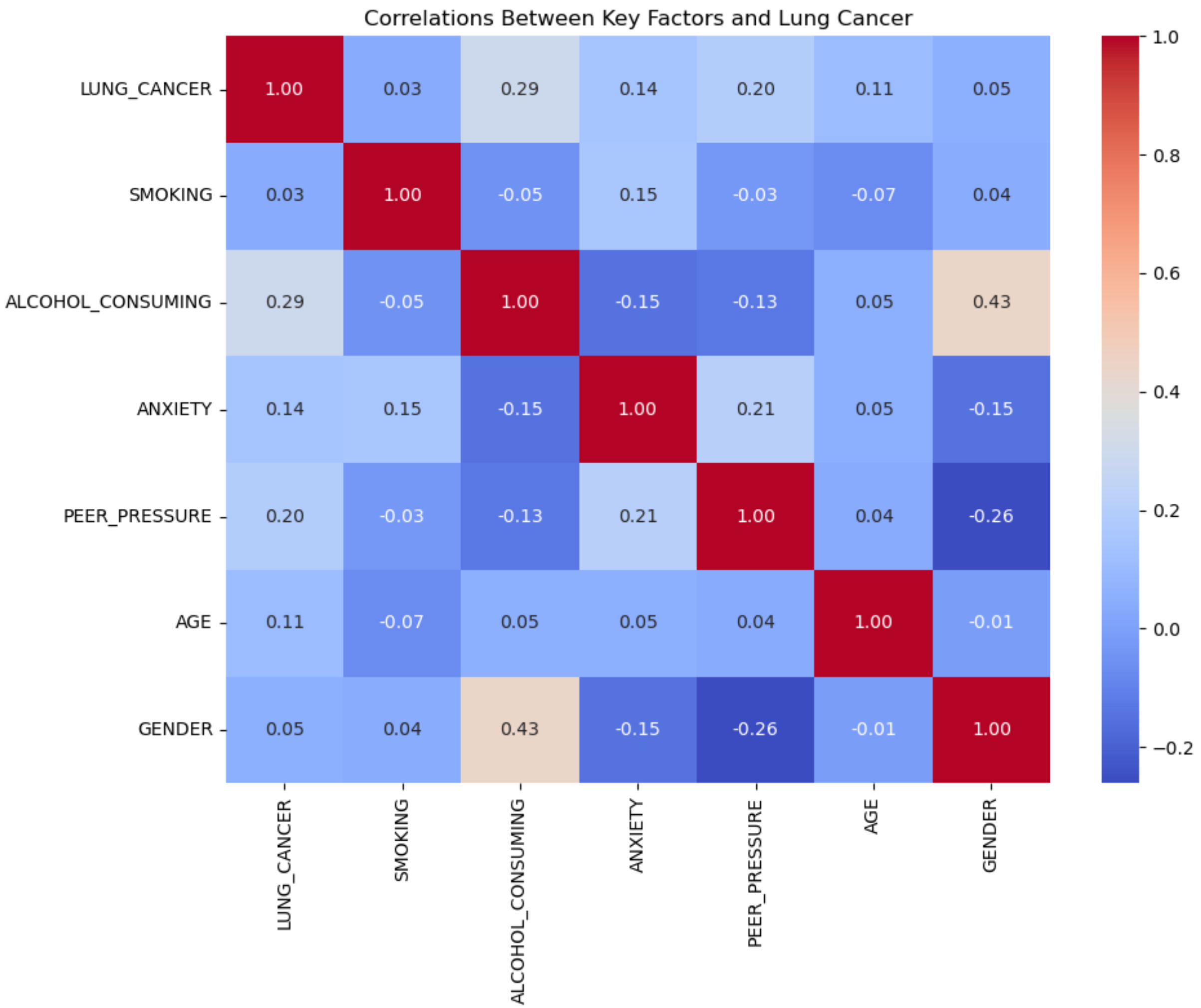
WEAK POSITIVE

Gender = 0.0536

Smoking = 0.0349

NO NEGATIVE

CORRELATION ANALYSIS



PSYCHOSOCIAL

Anxiety = 0.1443

Peer pressure = 0.1951

BEHAVIORAL

Smoking = 0.0349

Alcohol Consumption = 0.2944

Serviceable available market

MODERATORS

Age = 0.1063

Gender = 0.0537

RESULTS

MODEL COMPARISON

BASE

For null hypothesis

INTERACTION

Psychosocial and Behavioural influence.

FULL

Does age and gender moderate?

INFERENTIAL STATISTICS

LOGISTIC REGRESSION

AUC = 0.87, accuracy = 86%

LIKELIHOOD RATIO TESTS

Base vs Interactions: Chi2=13.24, df=4.0, p-value=0.0102

Interaction vs Full Model: Chi2=11.05, df=8.0, p-value=0.1991

WALDS TEST

ALCOHOL_CONSUMING

Other significant:
ALCOHOL_ANXIETY

PEER_PRESSURE

CONFIDENCE INTERVAL

Alcohol Consumption 65 x

(Coefficient: +4.17, p < 0.001)

Peer pressure 4 x

(Coefficient: +1.43, p = 0.027)

Anxiety moderates Alcohol

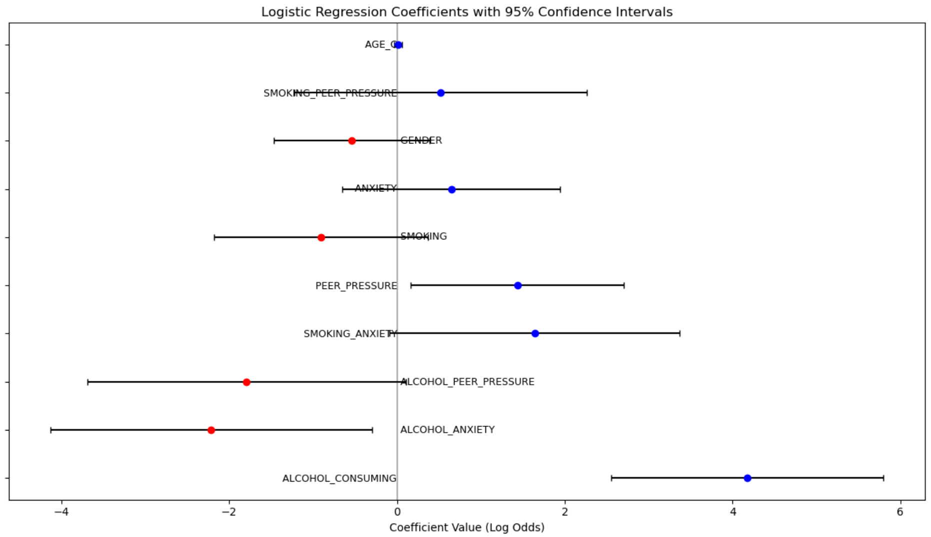
(Coefficient: -2.21, p = 0.024)

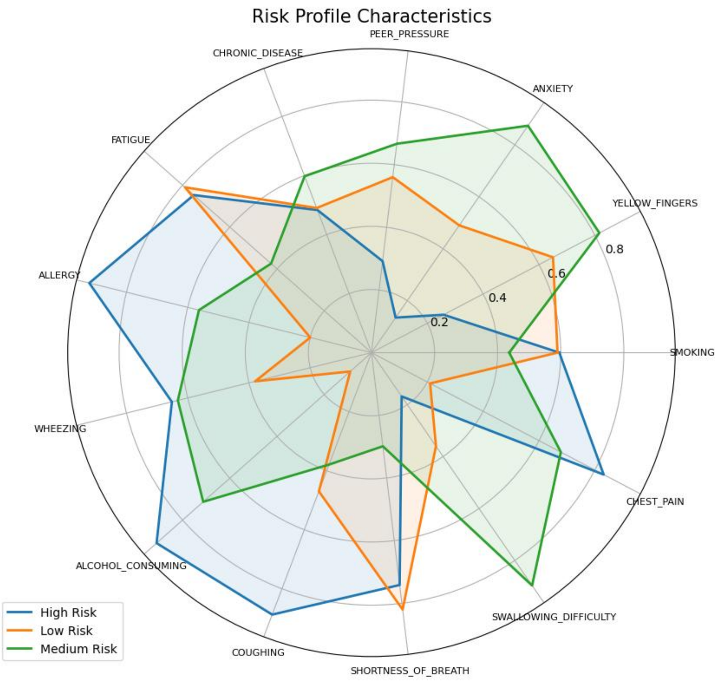
ANOVA TEST

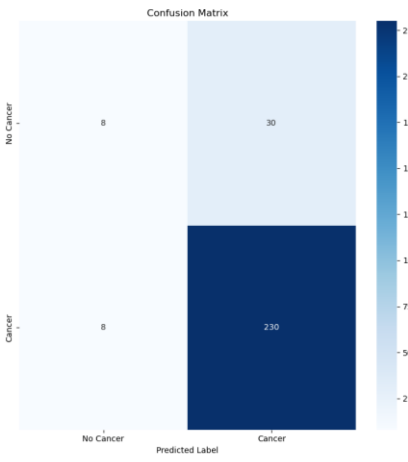
High risk level more likely

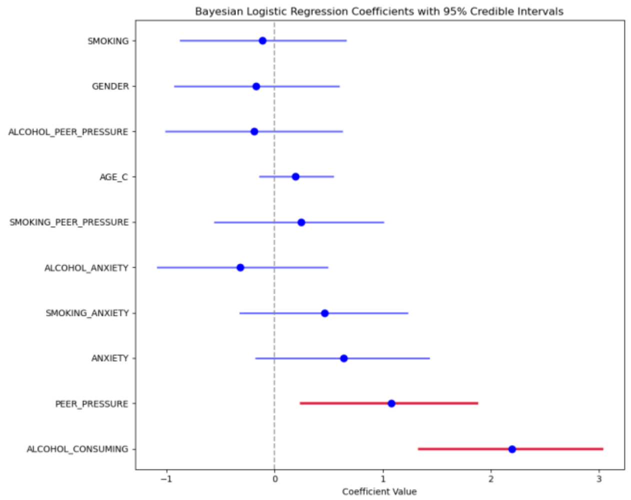
Gender is not significant

Smoking vs Alcohol (Alcohol or with Smoking)









BAYESIAN STATISTICS

BAYESIAN REGRESSION

AUC = 0.82, accuracy = 87%

WAIC TESTS

Base Model: -92.72

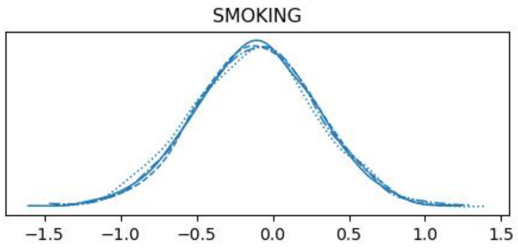
Interaction Model: -91.31

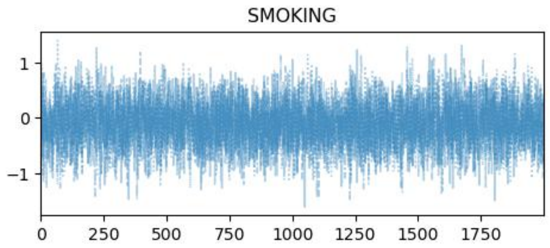
Full Model: -91.74

APPROXIMATE BAYES FACTORS

Interaction vs Base: 12270.90

Full vs Interaction: 2078844268.89





LOO TEST

interaction 0.000000

full 1.884306

base 1.823951

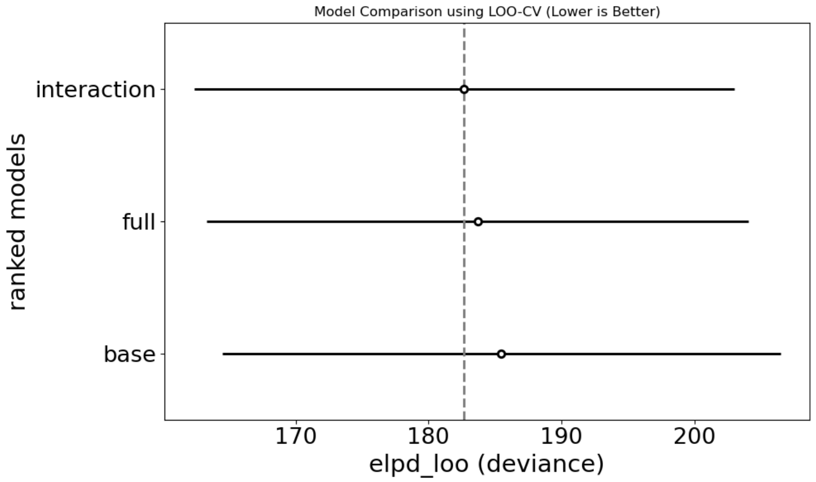
CREDIBLE INTERVAL

Alcohol Consumption 9.9 x

(OR: (3.75, 22.43))

Peer pressure 3.23 x

(OR: (1.25, 6.85))



SUMMARY

Analysis rejects null hypothesis.
Psychosocial and behavioural factors do in fact influence lung cancer.

THANK YOU

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